An AI-driven Stress Detection System using Physiological Signals

By

Ms.Krittaporn Ruknui Student ID: 65100372

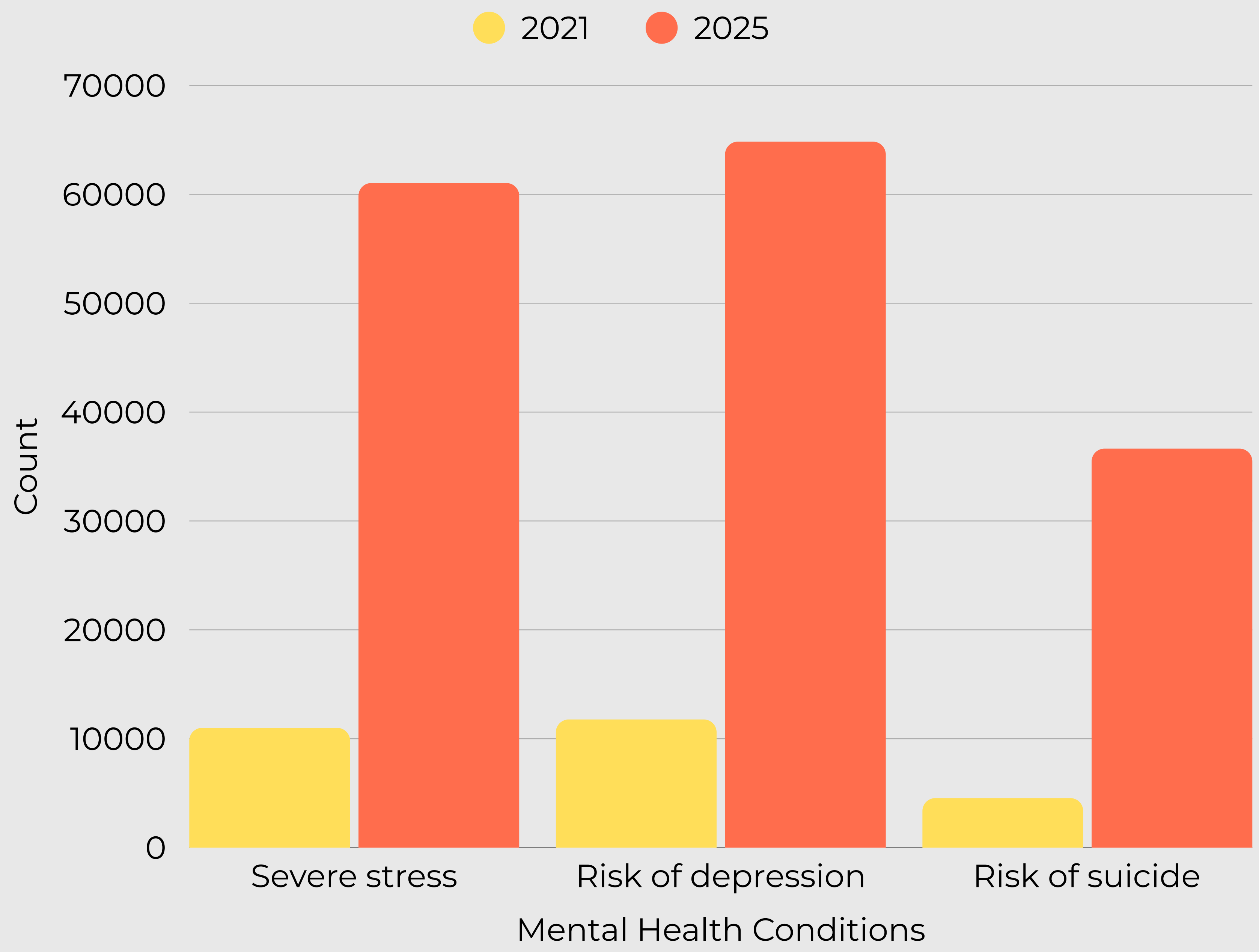
Mr.Surapat Hoknoo Student ID: 65122723

Mr.Surapat Saetan Student ID: 65123465

Advisor: Assistant Dr. Mallika Kliangkhlao

Computer Engineering and Artificial Intelligence
School of Engineering and Technology
Walailak University, Thailand

Mental Health Conditions: Comparison of 2021 and 2025



Severe Stress have shown a dramatic **increase**

454.5% 2021 to 2025

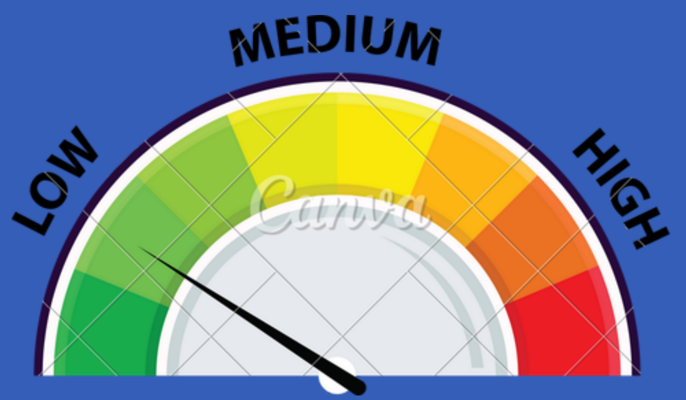
Health Data Repository, Department of Mental Health.
Data on mental health assessment of Thai people.
(Data as of August 15, 2025)

PHYSIOLOGICAL SENSORS

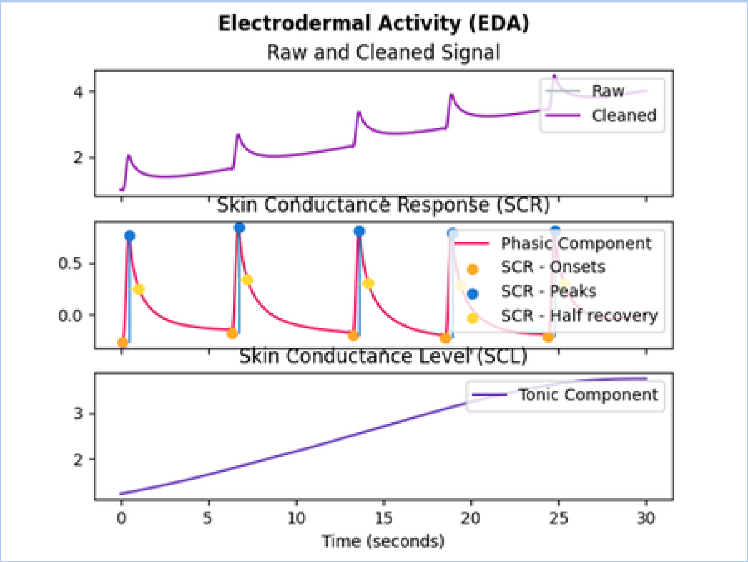
Autonomic Nervous System

SNS Sympathetic Nervous System

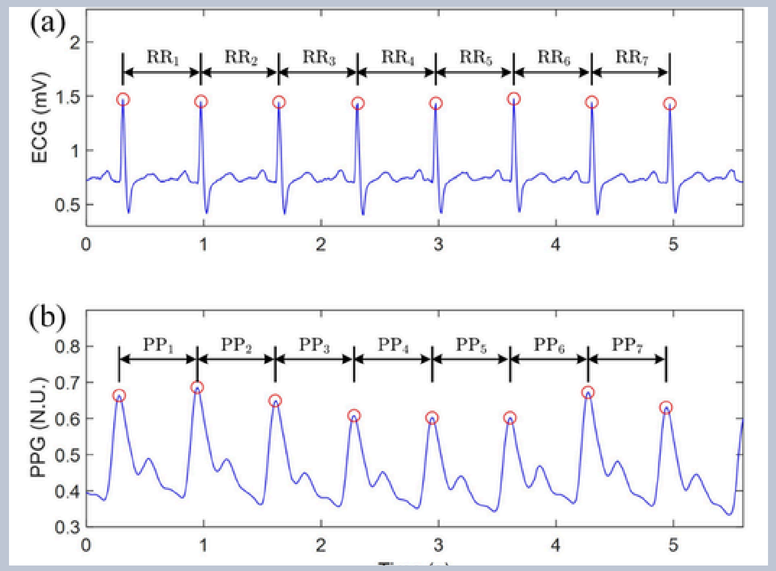
PNS Parasympathetic Nervous System



EDA Electrodermal Activity



HRV Heart Rate Variability



OBJECTIVES

Study the concept

To study the conceptual framework of applying artificial intelligence technology and the measurement of electrodermal activity (EDA) signals and heart rate variability (HRV) for stress detection.

Develop a model

To develop a learning model for stress level classification using electrodermal activity signals and heart rate variability.

Develop a system

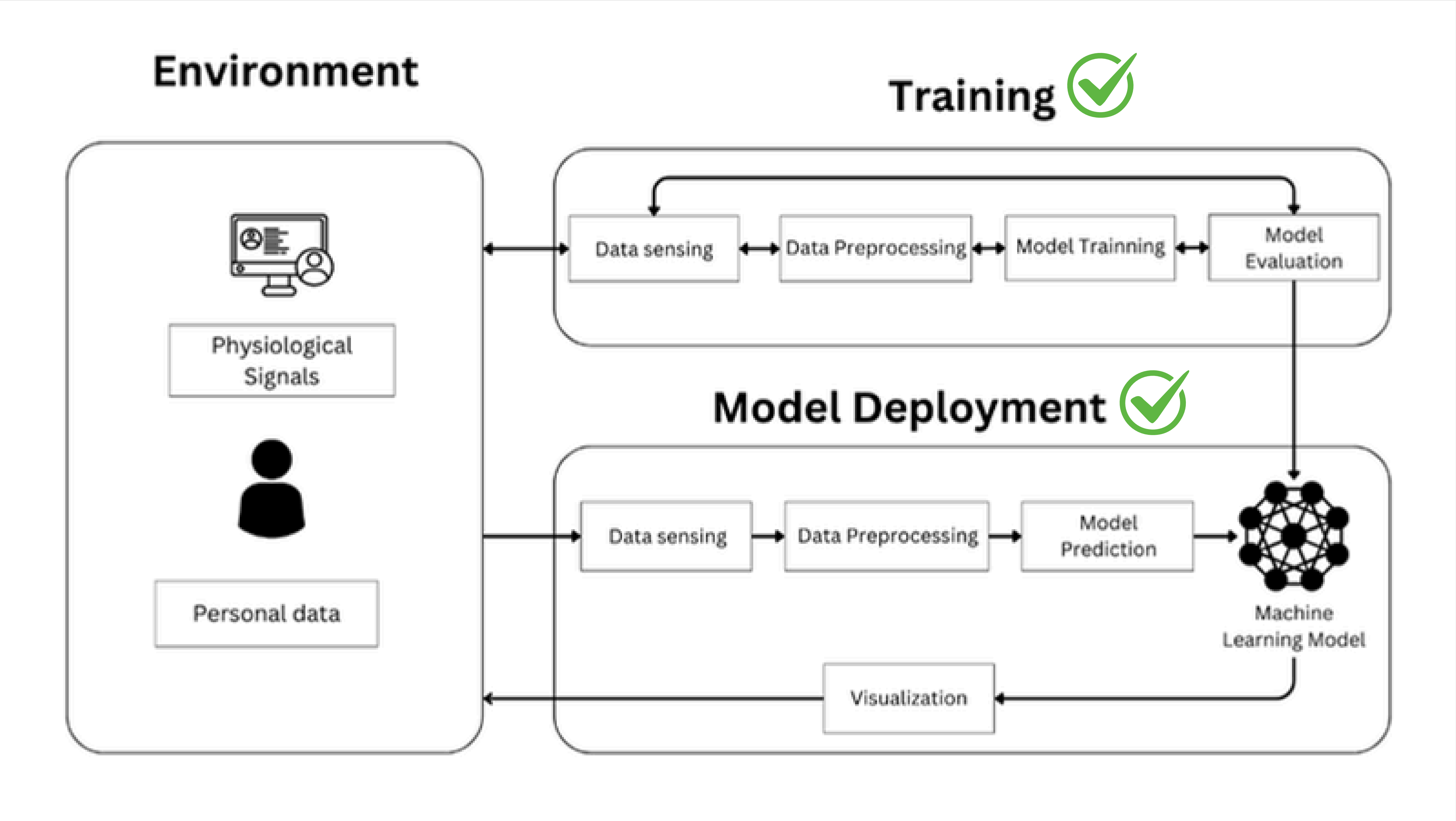
To develop a prototype system for real-time stress detection.

LITERATURE REVIEWS

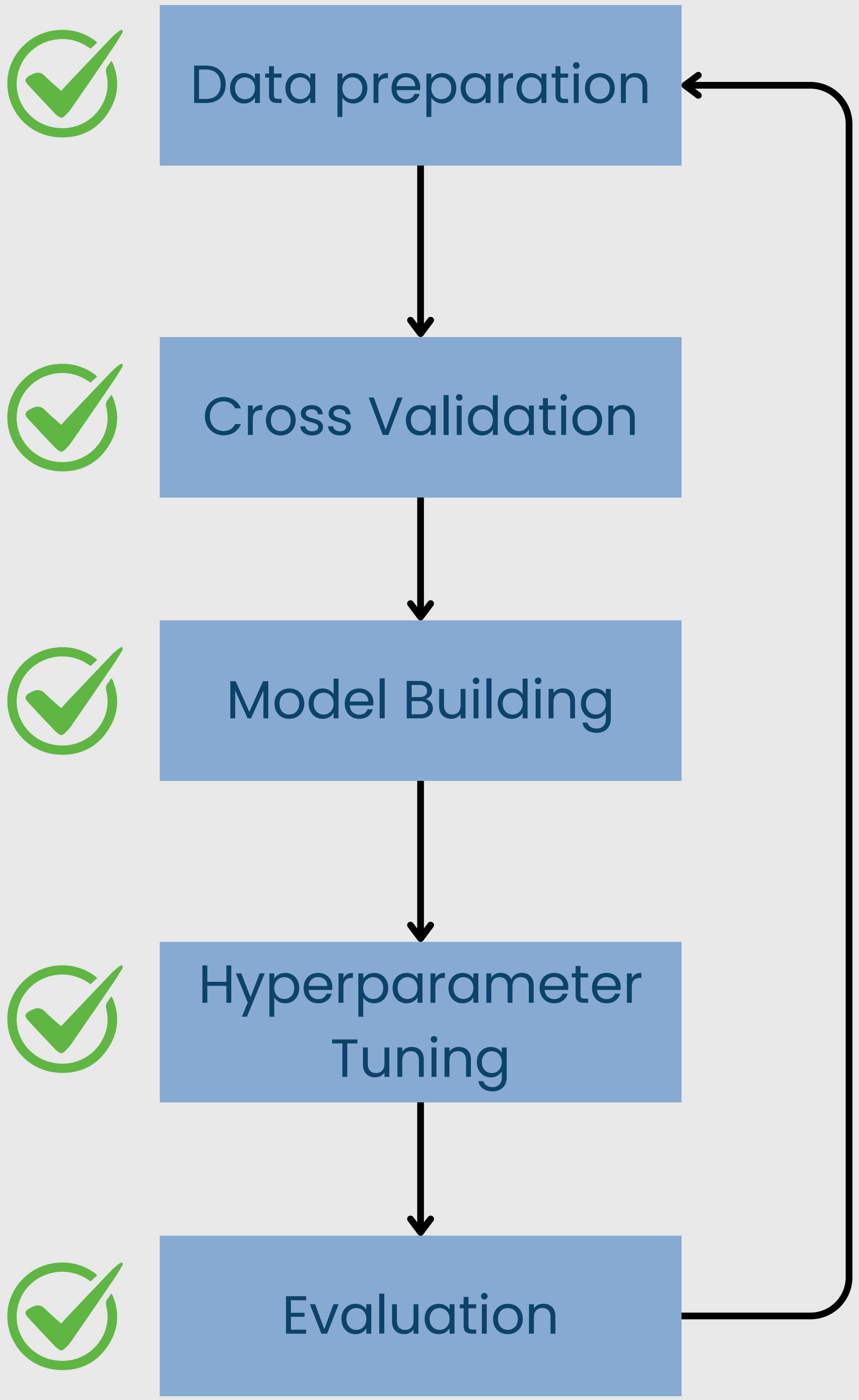
Comparison	The proposed innovation	[1]	[2]	[3]	[4]	[5]	[6]	[7]
Provide initial guidance	✓		-		✓	-	✓	✓
Use AI for detection	✓	✓	-		✓	-	✓	✓
Automatic notifications	✓		-	✓		-	✓	✓
Summary report of stress trends	✓		-	✓	✓	-	✓	✓
Signals collected	EDA and HRV	EDA and HRV	HRV	Cortisol in Sweat and HRV	-	-		
Usage characteristics	Wearable device installed on the wrist	Worn on the finger	Attached to various points on the body	Wearable device worn on the wrist (smartwatch)	-	Development of a Personal Thermal Comfort System	Installation of Wearable Devices	Monitoring of Sleeping Postures

FRAMEWORK

Framework of the stress monitoring system

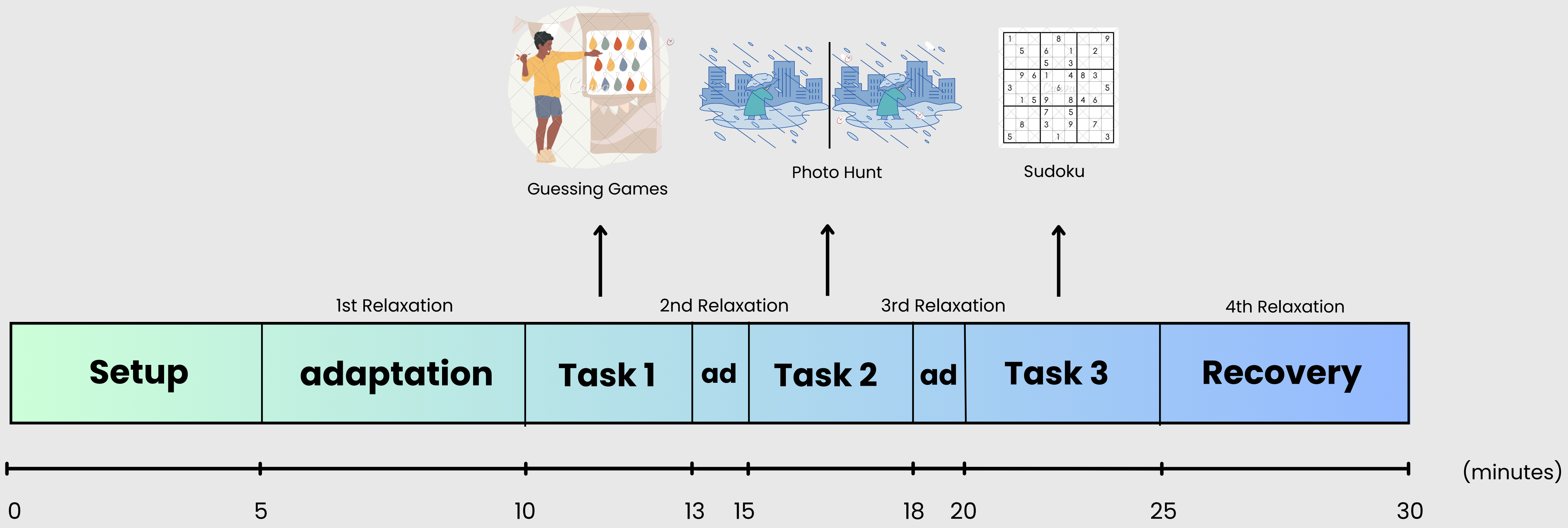


MODEL


Data preparation
Cross Validation
Model Building
Hyperparameter Tuning
Evaluation

DATA COLLECTION

3 Stress Tests



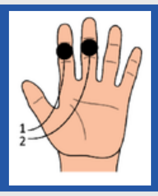
- Sample Group 25 participants
- Aged between 18-26 years
- No underlying conditions that severely affect the ANS
- History of severe psychiatric disorders
- Experiencing abnormally high stress levels at the time of the experiment

Human Research Project Code: WU-EC-EN-2-110-67

FEATURE ENGINEERING

Feature extraction from EDA and PPG signals can be performed using the **NeuroKit2** library, a Python library designed for analyzing physiological signals.

EMOTIBIT



EDA

- Tonic
- Phasic
- Amplitude
- Onsets

PPG

- **SDNN**: Standard Deviation of NN intervals
- **RMSSD**: Root Mean Square of Successive Differences
- **HRV_LF**: Low Frequency Power
- **HRV_HF**: High Frequency Power
- **HRV_LFHF**: LF/HF Ratio

MODEL DEVELOPMENT SETTING

PROBLEM

Our challenge involves analyzing time sequence data for stress detection using physiological measurements. The data consists of continuous temporal sequences from multiple sensors including ppg_rate requiring models that can effectively process and learn from sequential patterns.

ALGORITHM

It is necessary to measure the performance of several types of models

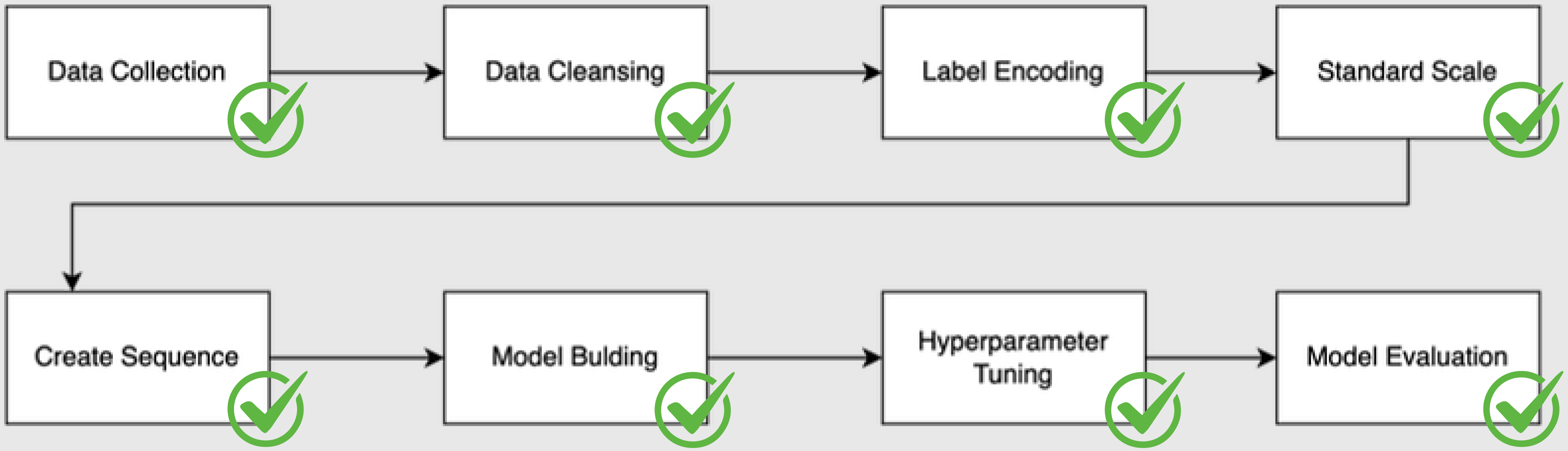
- RNN: Suitable for sequential data
- LSTM: enhance RNN
- GRU: offer a simplified and computationally efficient alternative to LSTMs while maintaining comparable performance on many sequence tasks.

MODEL DEVELOPMENT SETTING

EVALUATION METRICS

- Accuracy: measures the overall correctness of the model's predictions.
- Precision: indicates the proportion of true positive predictions among all positive predictions, reflecting how precise the model is.
- Recall: measures the proportion of true positives correctly identified among all actual positives, highlighting the model's ability to detect relevant cases.
- F1-score: is the harmonic mean of precision and recall, making it especially suitable for evaluating performance on imbalanced datasets.

PREPROCES SUMMARY



DATASET

- **Total participants**: 25 people
- **Total data points**: 3,252,775 values
- **Train/Test split**: 70:30 ratio

FEATURE SELECTION

Feature	Set Of Feature
EDA	EDA_Phasic, EDA_Tonic, SCR_Amplitude, SCR_Onsets, gender, bmi, sleep, Skintype, stress
HRV	HRV_RMSSD, HRV_LFHF, HRV_SDNN, HRV_LF, HRV_HF, gender, bmi, sleep, Skintype, stress
EDA + HRV	EDA_Phasic, SCR_Amplitude, EDA_Tonic, SCR_Onsets, HRV_RMSSD, HRV_LFHF, HRV_SDNN, HRV_LF, HRV_HF, gender, bmi, sleep, Skintype, stress

SET OF PARAMETERS

algorithm	parameter	value
RNN **LSTM** **GRU**	units	32-256 (step 32)
	dropout	0.05-0.2 (step 0.1)
	num_layers	1-5
	optimizer	adam, sgd
	learning_rate	1e-4 to 1e-2 (log scale)
	batch_size	8-128 (step 8)
	epochs	10-100

MODEL RESULT

EDA

Model	Hyperparameter							Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
	Neural Unit No	dropout	Hidden Layer No	optimizer	Learning rate	Batch size	epochs				
RNN	96	0.05	2	adam	0.0003	56	65	93.14	93.44	93.11	93.15
LSTM	256	0.2	2	adam	0.001	64	50	94.98	95	95	95.01
GRU	256	0.05	1	adam	0.000139	56	57	93.45	93.64	93.45	93.45

MODEL RESULT

HRV

Model	Hyperparameter							Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
	Neural Unit No	dropout	Hidden Layer No	optimizer	Learning rate	Batch size	epochs				
RNN	128	0.05	5	adam	0.0001	64	100	99.5	99.5	99.5	99.5
LSTM	128	0.1	2	adam	0.001	64	70	99.53	99	99	99.53
GRU	256	0.05	2	adam	0.000828	32	65	99.33	99	99.67	99.33

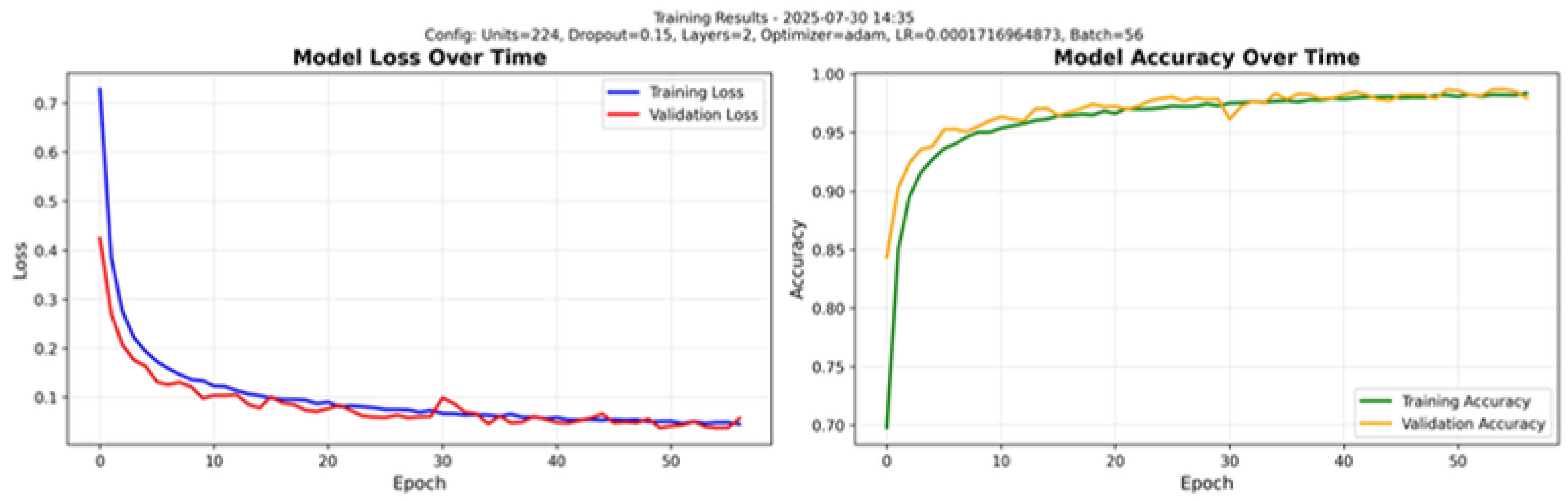
MODEL RESULT

EDA + HRV

Model	Hyperparameter							Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
	Neural Unit No	dropout	Hidden Layer No	optimizer	Learning rate	Batch size	epochs				
RNN	224	0.15	2	adam	0.00017	56	57	99.4	99.4	99.4	99.4
LSTM	160	0.05	1	adam	0.00041	128	80	99.56	99.5	99.5	99.5
GRU	192	0.15	3	adam	0.000451	64	37	99.42	99.42	99.42	99.42

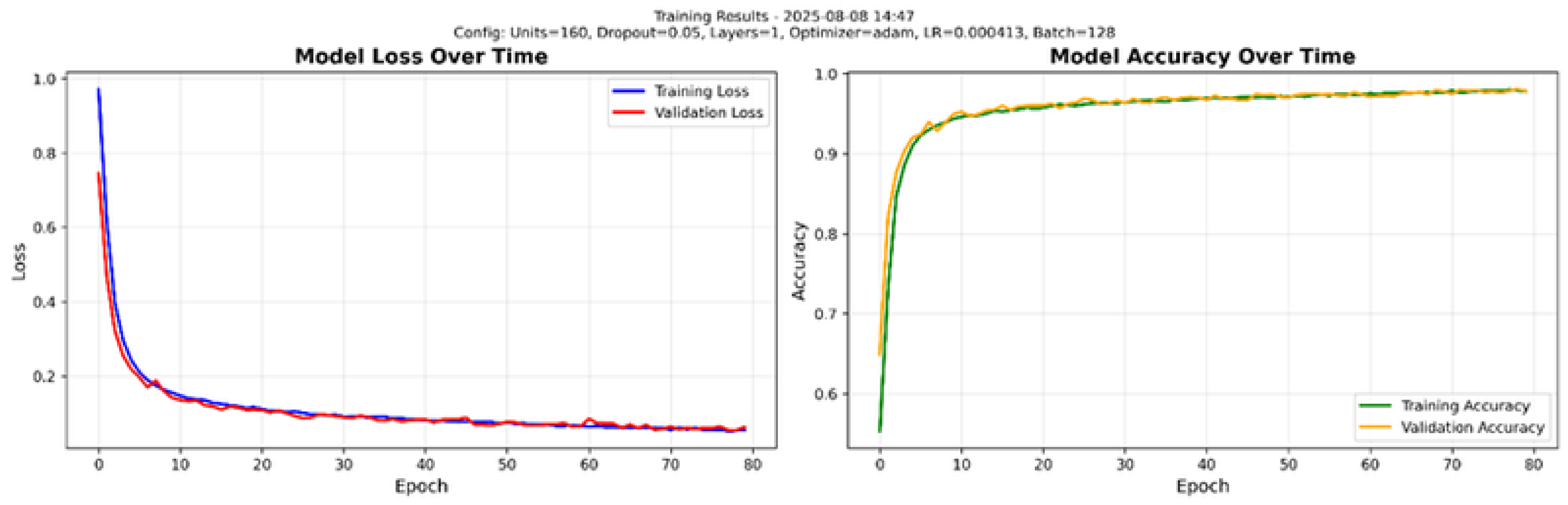
LOSS GRAPH

RNN



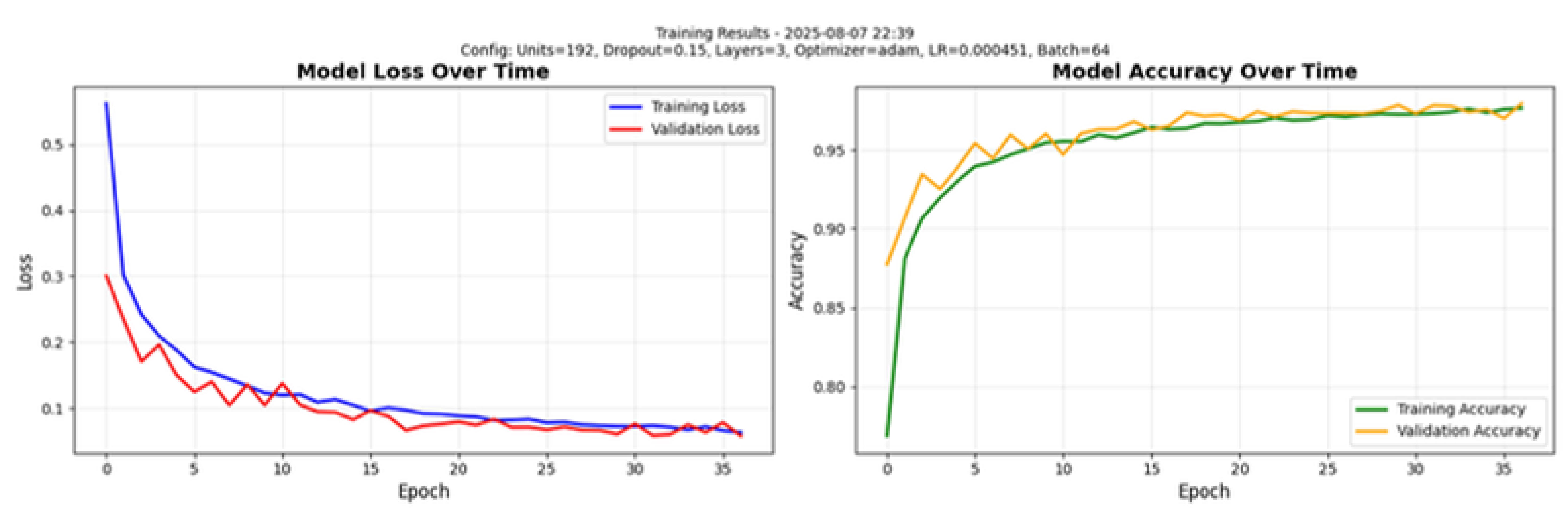
LOSS GRAPH

LSTM



LOSS GRAPH

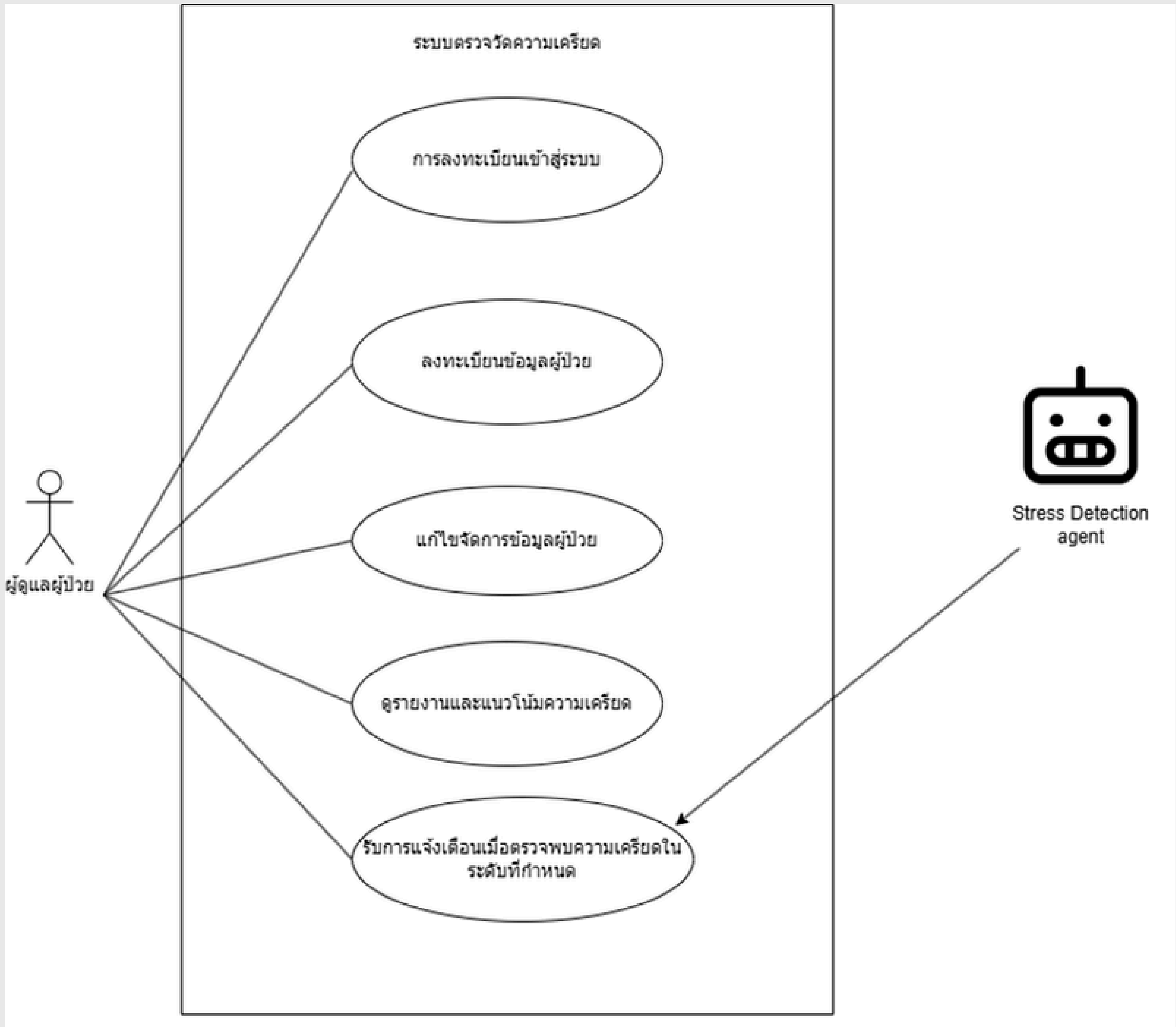
GRU



MODEL EVALUATION

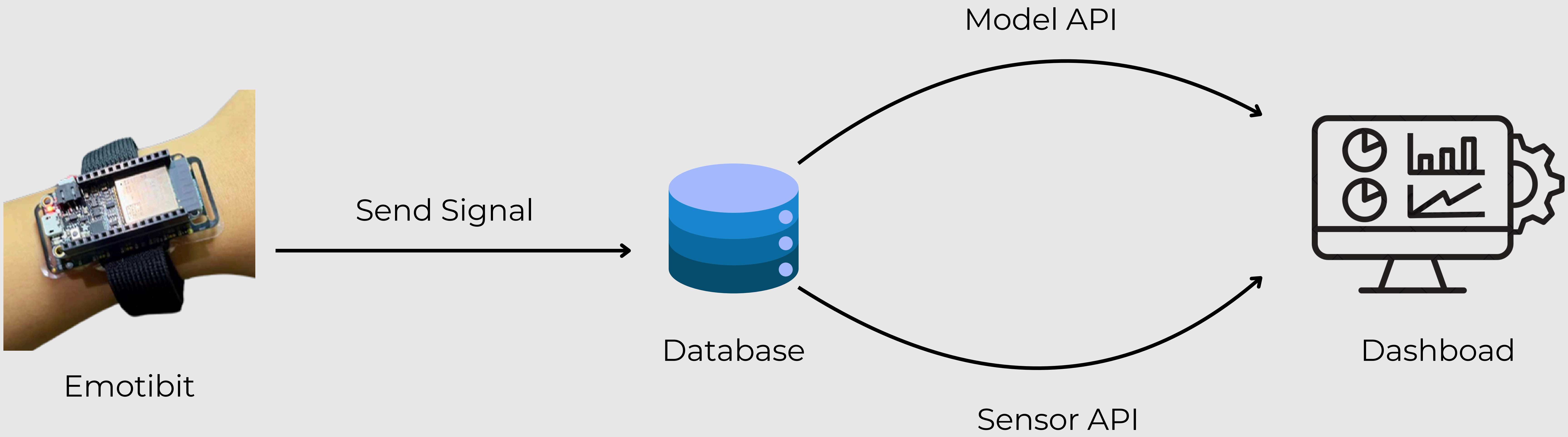
Model	Feature	Model Evaluation			
		Precision	Recall	F1-score	Accuracy
Recurrent Neural Network (RNN)	EDA	93.44	93.11	93.15	93.14
	HRV	99.5	99.5	99.5	99.5
	EDA + HRV	99.4	99.4	99.4	99.4
Long Short-Term Memory (LSTM)	EDA	95	95	95.01	94.98
	HRV	93	93	92.59	92.55
	EDA + HRV	97	97	97.27	97.24
Gated Recurrent Unit (GRU)	EDA	89.49	96.34	92.91	92.89
	HRV	95.73	98.54	97.14	97.15
	EDA + HRV	99.42	99.42	99.42	99.42

SCOPE OF WORK



- User management
- Register patient data
- Edit patient data
- Stress detection report
- Stress detection alert

SYSTEM FRAMWORK



DATABASE DESIGN

Device_status
_id
device_id
status
timestamp

Stress_prediction
_id
device_id
result
timestamp

Sensor_readings
_id
topic
device_id
recieve_at
timestamp
sensor

Preprocessed_data
_id
device_id
eda_feature
ppg_feature
hrv_indices
timestamp

Personal_data
_id
device_id
name
surname
age
weight
height
CD
SkinType
timeToSleep

INFRASTRUCTURE

FRONTEND SERVICE

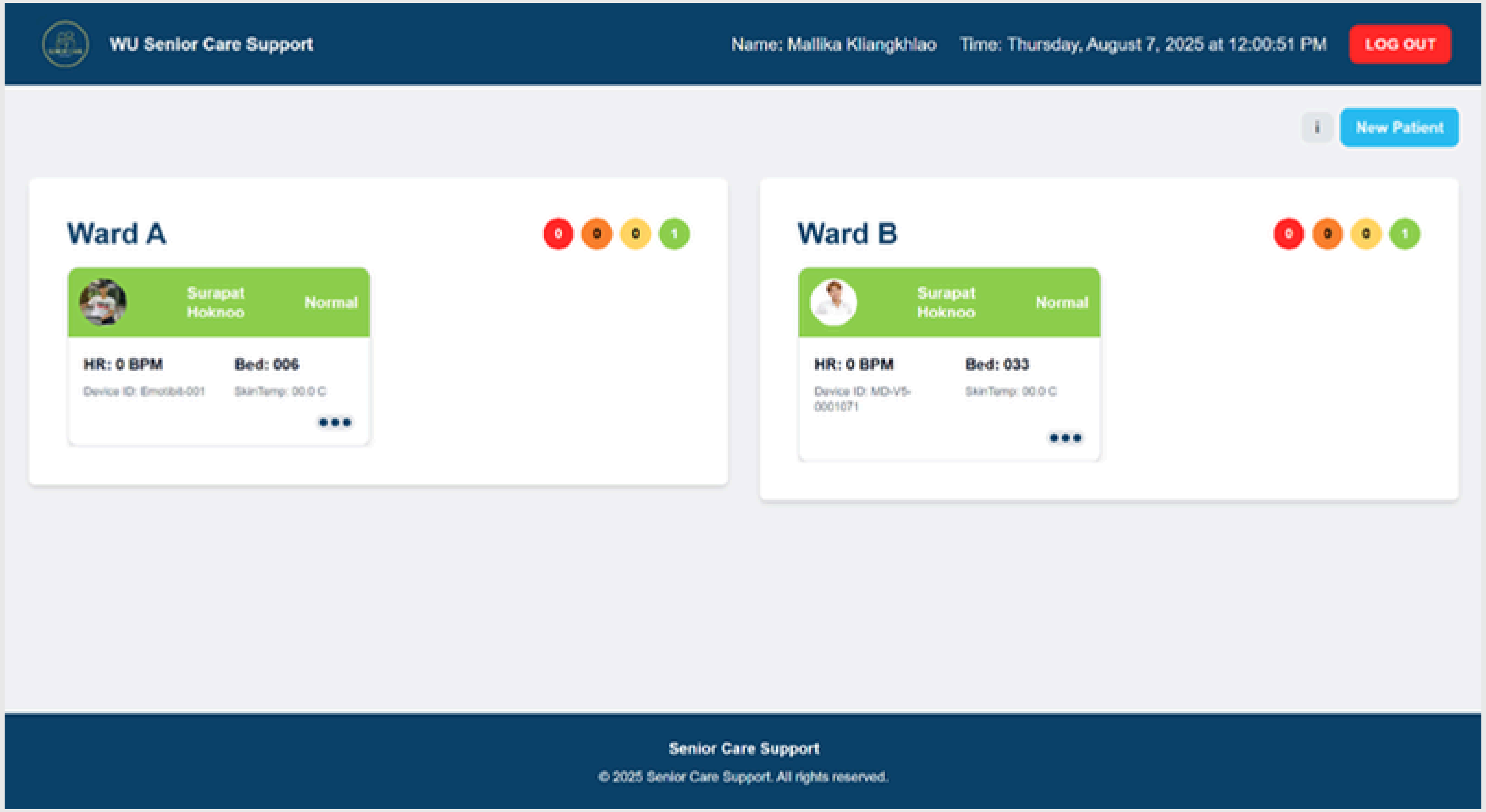
SERVER SERVICE

DATABASE SERVICE

WEB APPLICATION DEMONSTRATION

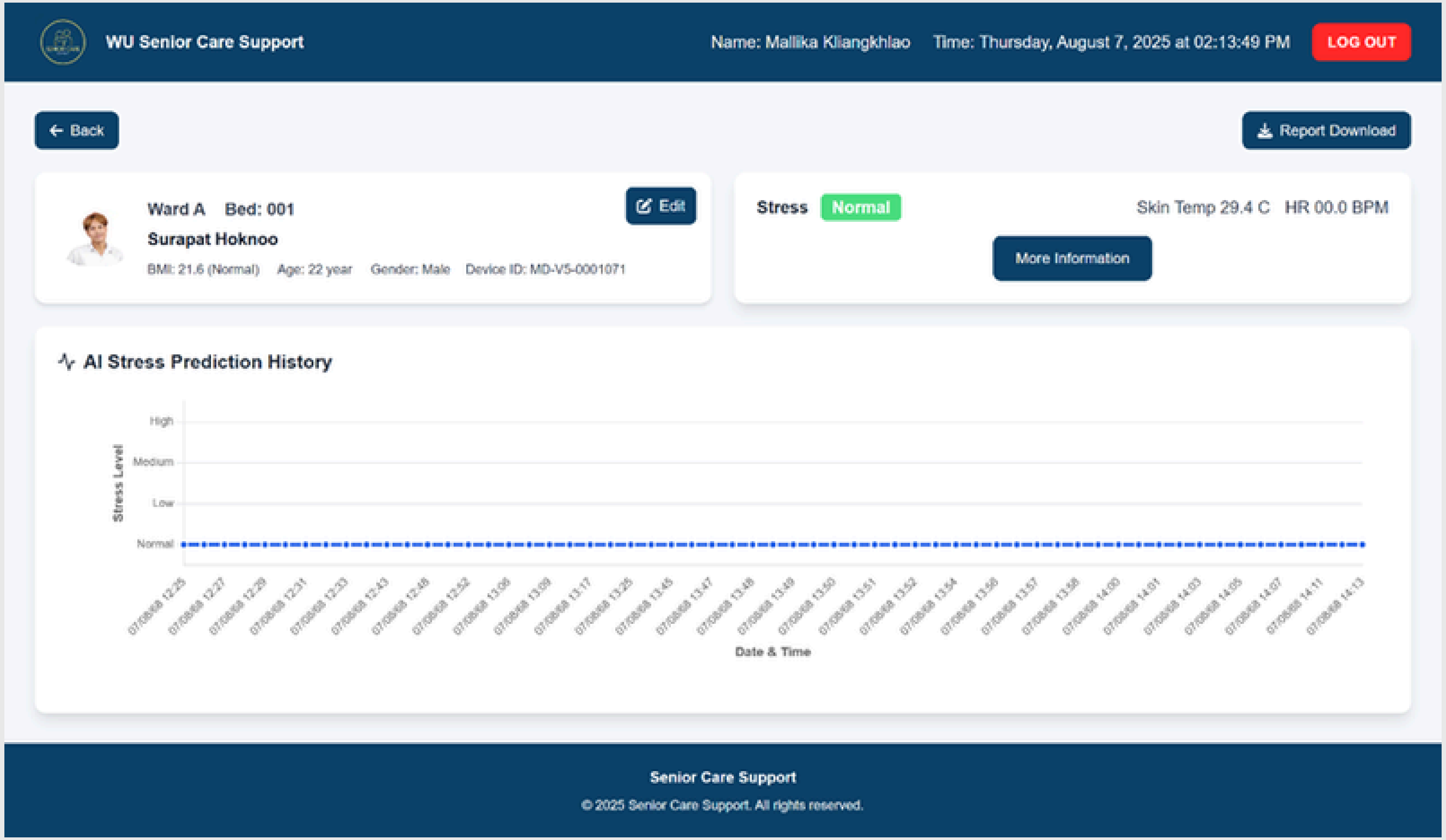
WEB APPLICATION

Dashboard Page



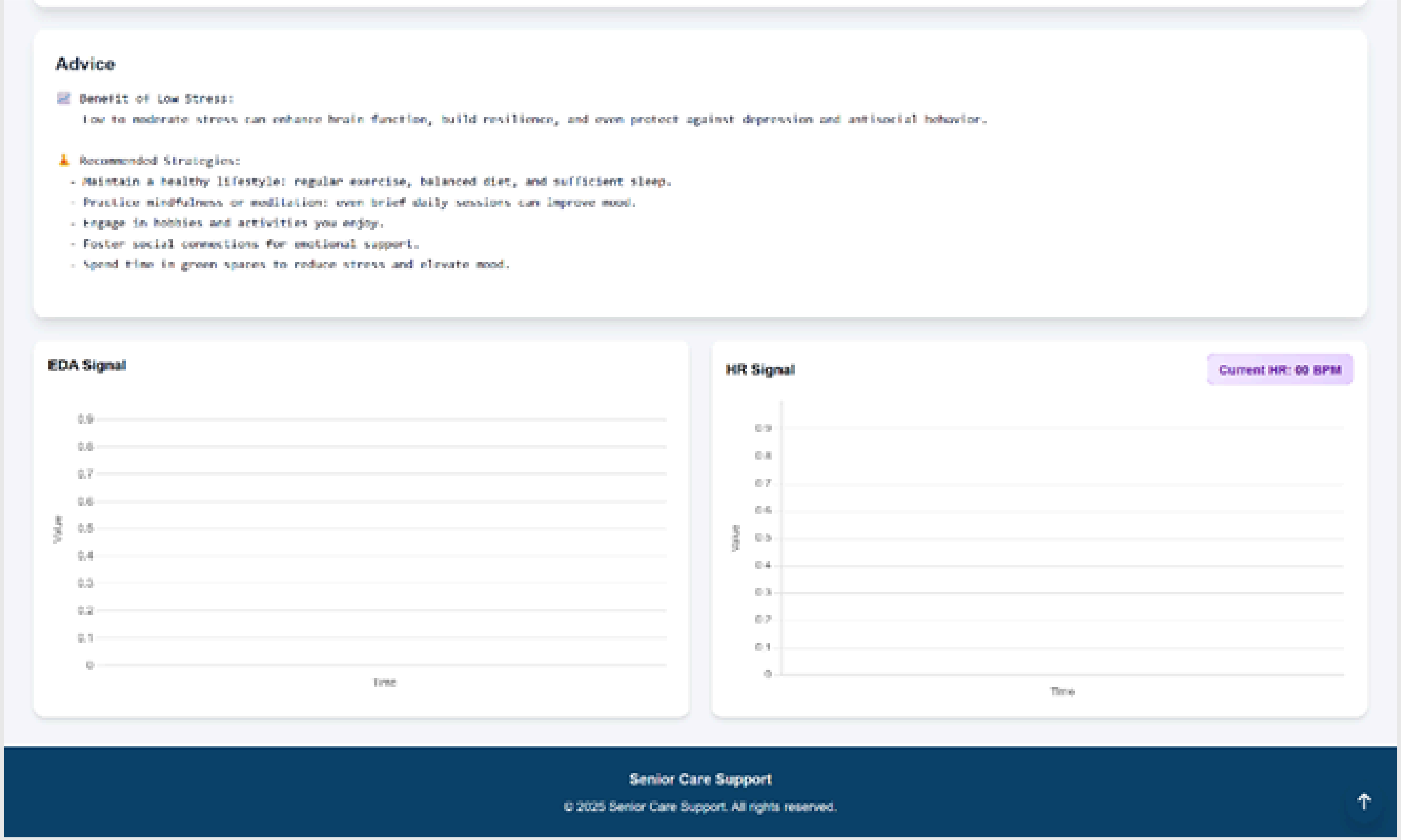
WEB APPLICATION

Detail Page



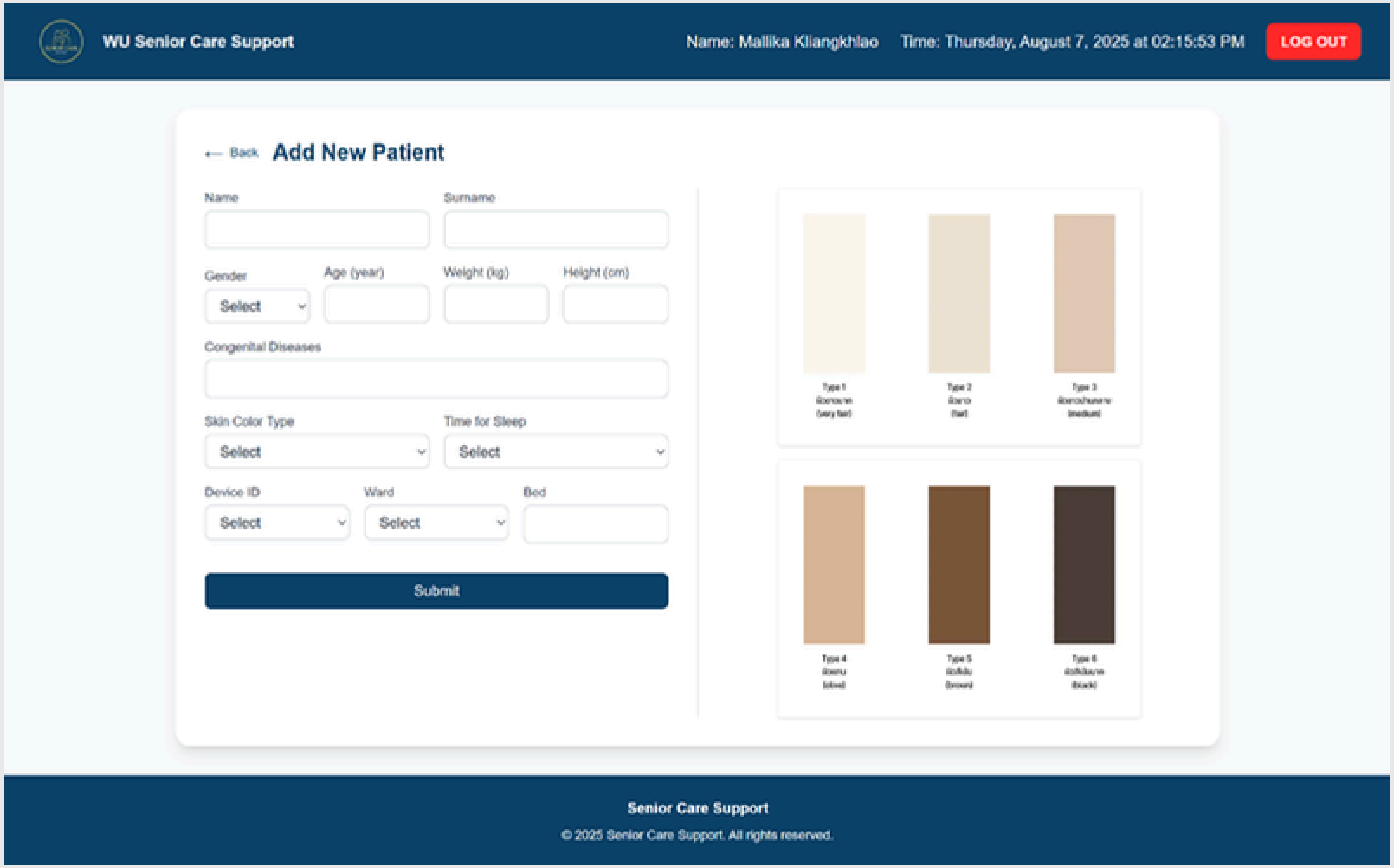
WEB APPLICATION

Detail Page



WEB APPLICATION

New Patient



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Q&A

THANK YOU FOR YOUR ATTENTION